Biology All In One (BAIO) Learning System

PU-II Biology · Narendra Sir – M.Sc (Biochemistry), B.Ed, KSET

Chapter 9: Biotechnology – Principles and Processes

Worksheet · Isolation of DNA & Process of RDT

I. Multiple Choice Questions

1. Which enzyme is used to break open a bacterial cell wall during DNA isolation?

   (A) Cellulase (B) Chitinase
   (C) Lysozyme (D) Protease

2. Which enzyme is used to break open a plant cell wall during DNA isolation?

   (A) Lysozyme (B) Cellulase
   (C) Chitinase (D) Ligase

3. Which enzyme is used to break open a fungal cell wall during DNA isolation?

   (A) Lysozyme (B) Cellulase
   (C) Chitinase (D) Protease

4. Which enzyme digests contaminating RNA during DNA isolation?

   (A) Protease (B) Ribonuclease
   (C) Ligase (D) Restriction enzyme

5. Which enzyme digests contaminating protein during DNA isolation?

   (A) Ribonuclease (B) Protease
   (C) Ligase (D) Polymerase

II. Fill in the Blanks

1. The enzyme used to break open a bacterial cell wall during DNA isolation is _______.
2. _______ is used to break open plant cell walls during DNA isolation.
3. Chitinase is used to break open _______ cell walls during DNA isolation.
4. _______ digests contaminating RNA, while _______ digests contaminating protein.
5. Chilled _______ is added at the final step to precipitate purified DNA.
6. After restriction digestion, DNA fragments are isolated using _______.

III. Match the Following

Column A	Column B
1. Lysozyme	a. Digests contaminating RNA
2. Cellulase	b. Breaks bacterial cell walls
3. Ribonuclease	c. Joins gene of interest with vector
4. DNA ligase	d. Breaks plant cell walls

IV. True / False

1. The same enzyme is used to break open bacterial, plant, and fungal cell walls. [True / False]

2. Ribonuclease and protease are used to remove RNA and protein contaminants respectively. [True / False]

3. Purified DNA appears as fine threads after ethanol precipitation. [True / False]

4. Gel electrophoresis is used to isolate DNA fragments after restriction digestion. [True / False]

5. DNA can be used for rDNA technology even in an impure form. [True / False]

V. 2 Marks Questions

1. Describe the sequence of steps involved in isolating pure DNA from a cell.

2. Why must DNA be obtained in pure form before it can be used in rDNA technology?

VI. 3 Marks Questions

1. What role does DNA ligase play in forming recombinant DNA?

2. Explain the purpose of adding chilled ethanol during DNA isolation.

VII. 5 Marks Questions

1. Describe, in the correct sequence, the steps involved in isolating and purifying DNA from a cell, mentioning the specific enzymes used to break open bacterial, plant, and fungal cell walls.